

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Supplementary End Semester Examination – Summer 2023
Branch: Mechanical Engineering

Semester: III

Course: B Tech
Subject Code & Name: Material Science and Metallurgy (BTMEC302)
Max Marks: 60
Duration: 3 Hr.
Date: 10/08/2023

Instructions to the Students:

1. Each question carries 12 marks.
 2. Attempt any Five questions of the following.
 3. Illustrate your answer with neat sketches, diagram, etc., wherever necessary.
 4. Assume suitable data wherever necessary & mention it clearly.
- (Marks)**

Q.1 Solve the following.

- a) Explain in short different imperfections in crystal structure. (12)
- b) Explain slip mechanism of plastic deformation. Give difference between slip and twinning. 6

Q.2 Solve the following.

- a) Define Creep. Explain effect of creep rate on creep curve. 6
- b) Explain Brinell hardness test w.r.t. principle of working, indenter details, formula, advantages and limitations. 6

Q.3 Solve the following.

- a) Draw the Fe-Fe₃C diagram. Explain Phases and Critical Temperatures. (12)
- b) What is the importance of TTT diagram? Explain TTT diagram in detail. 6

Q.4 Solve the following.

- a) Define heat treatment and give its objectives. Explain tempering process w.r.t. types, variation of properties with tempering temperature. 6
- b) What is surface hardening? Explain induction hardening with neat sketch. 6

Q.5 Solve the following.

- a) Explain specimen preparation for optical metallurgical microscope. (12)
- b) Explain sulphur print test w.r.t. purpose, significance, and procedure. 6

Q.6 Solve the following.

- a) Give Different types of NDT. Explain in brief Magnaflux Test. 6
- b) Write short notes on: 6
 - 1) Basic Types of Crystal Structure
 - 2) Space Lattice and Lattice Parameters

End